Jonathan Dambros

Ph.D. in process control · Ph.D.-level knowledge to break the LLMs of the leading AI labs

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OVERVIEW

- **LLM evaluation.** Write expert-level tasks that are fair and that frontier models still fail, and lead a team of 30+ doing the same, across many projects for several of the world's largest AI labs.
- **Researcher.** 10+ years in process control and monitoring at UFRGS (GIMSCOP). 20+ publications, 300+ citations.
- **Programmer.** 10+ years of Python: scientific libraries, data pipelines and web tools running in industry.

SKILLS

- **Soft:** problem solving · attention to detail · domain knowledge · sustained deep focus · initiative · organized, well-documented work
- **Hard:** Python (10+ years) · machine learning (8+) · AI tools, used heavily and deliberately in daily work (4 years) · process control and control-loop monitoring (9+) · state estimation, Kalman/LQR (7+) · optimization (8+)

RESEARCH

- Oscillation and valve stiction detection in control loops
- State estimation with Kalman filters
- Predictive maintenance
- Machine learning for process monitoring
- Applied to refineries and offshore oil and gas, mainly with Petrobras

EXPERIENCE

Engineering Manager — LLM Evaluation, Turing — 2023 – Present

- Write expert-level control engineering tasks that frontier LLMs still fail, for several leading AI labs.
- Lead a team of 30+ reviewers evaluating LLM outputs with calibrated rubrics.
- Designed evaluation frameworks and reviewer workflows for reliability and consistency.

Postdoctoral Researcher, UFRGS — GIMSCOP — 2019 – Present

- Developed Python libraries (DigitalTwin, Kalman) in collaboration with Petrobras.
- Researched and deployed monitoring and diagnostics systems in process industries.
- Published peer-reviewed work on control-loop performance monitoring and optimization.

Team Lead (Self-employed), Latos — 2020 – 2023

- Built data-driven solutions for manufacturing and process industries.
- Developed and deployed “Heron,” a web platform for data science workflows.
- Integrated multi-source pipelines with visualization in Grafana, Power BI, Tableau.

SELECTED PROJECTS

- **awesome-industrial-datasets** — Curated collection of public industrial datasets, with ~600 GitHub stars.

EDUCATION

Postdoctoral Researcher — Chemical Engineering, UFRGS — 2019 – Present

Ph.D. — Chemical Engineering, UFRGS — 2016 – 2019

Visiting Researcher — Artificial Intelligence Group, University of Kaiserslautern, Germany — 2018

M.Sc. — Chemical Engineering, UFRGS — 2015 – 2016

B.Sc. — Chemical Engineering, FURG — 2010 – 2014

AWARDS

- Top Graduate, Chemical Engineering (2014)
- International Challenge Winner — Computer Vision for Scrap Inspection (2019)
- Best Industry 4.0 Application — National Prize (2021)